(d) $\dot{s} = \frac{1}{2} E(s)\, \delta\omega$ — (1)

$$J \delta\dot{\omega} = -\omega \times J\omega + u - J\left(\dot{R}_s \omega_d + R_s \dot{\omega}_d\right) \quad \text{— (2)}$$

First we assume $\delta\omega$ is the control. Expanding (1),

$$\dot{s}_0 = -\frac{1}{2} s_v^T \delta\omega$$
$$\dot{s}_v = \frac{1}{2}\left(s_0 I + S(s_v)\right)\delta\omega$$

Consider $V_1 = \|s_v\|^2 + (s_0+1)^2$ ---- {assuming $s_0 \to -1$ when control achieved}

$= 2(1+s_0)$ --- {since $\|s\| = 1$}

$$\dot{V}_1 = 2\dot{s}_0 = -s_v^T \delta\omega$$

If $\delta\omega = c_1 s_v$, $\dot{V}_1 = -c_1 \|s_v\|^2 \leq 0$ where $c_1 > 0$

But $\delta\omega$ is not the control,

define $z_1 = \delta\omega - c_1 s_v$

$$V = V_1 + \frac{1}{2} z_1^T J z_1$$

$$\Rightarrow \dot{V} = -s_v^T (z_1 + c_1 s_v) + z_1^T J \dot{z}_1$$
$$= -c_1 \|s_v\|^2 - s_v^T z_1 + z_1^T \left(-\omega \times J\omega + u - J\phi - c_1 \frac{J}{2}\left(s_0 I + S(s_v)\right)\delta\omega\right)$$
$$= -c_1 \|s_v\|^2 + z_1^T \left(u - \omega \times J\omega - J\phi - c_1 \frac{J}{2}\left(s_0 I + S(s_v)\right)\delta\omega - s_v\right)$$

We choose $u = \omega \times J\omega + J\phi + \frac{c_1}{2} J\left(s_0 I + S(s_v)\right)\delta\omega + s_v - c_2 z_1$

So that $\dot{V} = -c_1 \|s_v\|^2 - c_2 \|z_1\|^2 \leq 0$

Barbalat's Lemma and Signal Chasing give as $t \to \infty$

$\underline{s_v \to 0, \; z_1 \to 0, \; \delta\omega \to 0}$.

$\to$ & $\underline{s_0 \to -1}$

(e) Adaptive Controller

$$\theta^* = \begin{bmatrix} J_{11} & J_{22} & J_{33} & J_{23} & J_{13} & J_{12} \end{bmatrix}^T$$ since J is symmetric.

$$-\omega \times J\omega - J\phi = -S(\omega) J\omega - J\phi$$

For any $v \in \mathbb{R}^{3\times 1}$, $Jv = L(v)\, \theta^*$

where $L(v) = \begin{bmatrix} v[0] & 0 & 0 & 0 & v[2] & v[1] \\ 0 & v[1] & 0 & v[2] & 0 & v[0] \\ 0 & 0 & v[2] & v[1] & v[0] & 0 \end{bmatrix}$

Thus, $-\omega \times J\omega - J\phi = W\theta^*$

where $W = -S(\omega) L(\omega) - L(\phi)$

Our previous backstepping control law now becomes

$$u = -W\theta^* + \underbrace{\frac{c_1}{2} J\left\{\left(s_0 I + S(s_v)\right)\delta\omega\right\}}_{\frac{c_1}{2} L(a)\, \theta^*} + s_v - c_2 z_1$$

where $a = \left\{s_0 I + S(s_v)\right\}\delta\omega$

Thus, $u = W_1 \theta^* + s_v - c_2 z_1$

where $W_1 = \frac{c_1}{2} L(a) - W$

Since θ^* is not known, replace by $\hat{\theta}$

$$u = W_1 \hat{\theta} + s_v - c_2 z_1$$

Introduce a new term into the Lyapunov function,

$$V = 2(1+s_0) + \frac{1}{2} z_1^T J z_1 + \frac{1}{2} \tilde{\theta}^T \Gamma^{-1} \tilde{\theta}$$

Re iterating all the relevant equations from above,

$$\dot{S}_0 = -\frac{1}{2} S_v^T \delta\omega = -\frac{1}{2} S_v^T (z_1 + c_1 S_v)$$

$$z_1 = \delta\omega - c_1 S_v \quad , \quad J\delta\dot{\omega} = W\theta^* + u \; , \; \dot{S}_v = \frac{1}{2}(S_0 I + S(S_v))\delta\omega$$

$$\tilde{\theta} = \theta^* - \hat{\theta}$$

Thus, $\dot{V} = 2\dot{S}_0 + z_1^T J \dot{z}_1 - \tilde{\theta}^T \Gamma^{-1} \dot{\hat{\theta}}$ $\rightarrow - W_1\theta^*$

$$= -S_v^T (z_1 + c_1 S_v) + z_1^T \left(W\theta^* + W_1\hat{\theta} + S_v - c_2 z_1 - \frac{c_1 L(q)}{2}\theta^* \right) - \tilde{\theta}^T \Gamma^{-1} \dot{\hat{\theta}}$$

$$= -c_1 \|S_v\|^2 - \cancel{S_v^T z_1} + \cancel{S_v^T z_1} - c_2 \|z_1\|^2 + z_1^T (- W_1 \tilde{\theta}) - \tilde{\theta}^T \Gamma^{-1} \dot{\hat{\theta}}$$

$$= -c_1 \|S_v\|^2 - c_2 \|z_1\|^2 - \tilde{\theta}^T \left(W_1^T z_1 + \Gamma^{-1} \dot{\hat{\theta}} \right)$$

we choose $\underline{\dot{\hat{\theta}} = -\Gamma W_1^T z_1}$

(f) $\dot{s} = \frac{1}{2} E(s) \delta\omega$ — (1)

$J\delta\dot{\omega} = -\omega \times J\omega + \nu - J\phi$ — (2)

$\dot{\nu} = u$ — (3)

One more back-stepping !!!

Same as above,

$$z_1 = \delta\omega - c_1 S_v$$

$$J\dot{z}_1 = - W_1\theta^* + \nu$$

Lyapunov function $V = 2(1+S_0) + \frac{1}{2} z_1^T J z_1 + \frac{1}{2}\tilde{\theta}^T \Gamma^{-1}\tilde{\theta}$

gives

$$\dot{V} = -S_v^T (z_1 + c_1 S_v) + z_1^T (-W_1\theta^* + \nu) - \tilde{\theta}^T \Gamma^{-1} \dot{\hat{\theta}}$$

use the same control as above (assuming ν is the control)

$$\alpha = W_1\hat{\theta} + S_v - c_2 z_1$$

& introduce $z_2 = \nu - (W_1\hat{\theta} + S_v - c_2 z_1) = \nu - \alpha$

$$\Rightarrow \quad \dot{z}_2 = u - \dot{W}_1 \hat{\theta} - W_1 \dot{\hat{\theta}} + \dot{S}_v - c_2 \dot{z}_1$$

$$\dot{V} = - c_1\|S_v\|^2 - c_2 \|z_1\|^2 + z_1^T (z_2 - W_1 \tilde{\theta}) - \tilde{\theta}^T \Gamma^{-1} \dot{\hat{\theta}}$$

Introduce another term $\frac{1}{2} z_2^T z_2$ to the Lyapunov function

so that

$$\dot{V} = -c_1\|S_v\|^2 - c_2\|z_1\|^2 + z_1^T z_2 - \tilde{\theta}^T \left(W_1^T z_1 + \Gamma^{-1}\dot{\hat{\theta}} \right) + z_2^T \dot{z}_2$$

To handle the $\tilde{\theta}^T$ term, we choose

$$\dot{\hat{\theta}} = - \Gamma W_1^T z_1 \quad \text{same as above}$$

So now we are left with

$$z_1^T z_2 + z_2^T \dot{z}_2 = z_1^T z_2 + z_2^T (u - \dot{\alpha})$$

where α is our virtual control

Now we can choose u to be

$$\underline{u = \dot{\alpha} - z_1 - c_3 z_2}$$

This would result in z_1, z_2 terms cancelling out,

and we are left with

$$\dot{V} = - c_1\|S_v\|^2 - c_2 \|z_1\|^2 - c_3 \|z_2\|^2$$

$$u = \dot{\alpha} - z_1 - c_3 z_2$$

$$\underline{\dot{\hat{\theta}} = - \Gamma W_1^T z_1}$$

NOTE:

I have calculated α using finite difference in my implementation (because while calculating it analytically, it was way toooo long and wasn't sure if it can be even done because even without using the parameter estimate and differentiating everything in α, the results were not linear in θ and thus could not factor out $\tilde{\theta}$ from there to get a new update law)